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# Size, Shape, Stability, and Color of Plasmonic Silver Nanoparticles

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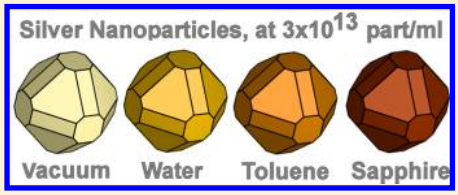
**ABSTRACT:** The remarkable relationship between the size, shape, and optical properties of silver nanoparticles is proving to be very useful in a range of high performance applications. Considerable effort and investment are focused on delivering silver nanoparticles with precise morphologies. However, the reliability of these particles is contingent upon the morphological stability, particularly against variations in the thermodynamic environment, such as changes in temperature. Using a combination of computational and theoretical approaches, we have constructed a size- and shape-dependent phase diagram of nanoscale silver and projected the optical emission spectrum to produce a detailed structure/property map at thermal equilibrium. This map demonstrates that faceted near-spherical shapes and spheres are the predominant morphologies in a Ag NPs colloidal sample at low and high temperatures, showing a light yellow color. However, samples with more faceted shapes such as cubes or tetrahedrons, which gain stability at sizes smaller than 8 nm at intermediate temperatures, will present brighter yellow hues and dark red colors, respectively.




## ■ INTRODUCTION

The ability to synthesize metal nanoparticles (NPs) with specific sizes and shapes based on a limited set of process parameters is an important area of research, as the control in the morphology is critical to our understanding and anticipating the properties and behavior under different conditions.[1−4] However, the production of well-defined NPs with reproducible size and shape distributions remains challenging.[5−11] When these challenges are overcome, metal NPs are used in a range of applications from catalysis[12] to optical and biological detection.[13−16]

Among the metal NPs being produced today, silver NPs are of great interest due to their potential use as optical labels,[17,18] contrast enhancement agents,[19,20] chemical and biological sensors,[21−24] and substrates for surface-enhanced Raman spectroscopy (SERS).[25,26] These applications are possible because silver NPs have novel optical properties originating from coherent oscillation of the conduction electrons upon exposure to light impinges of specific wavelength, known as the localized surface plasmon resonance (LSPR). This gives rise to enhanced light scattering and absorption, and large local field enhancement, near the NP surface at the resonant condition,[27] which can be controlled by modifying the size, shape, composition,[28−32] and dielectric environment.[6,33]

Considerable attention has been given to understanding the relationship between the optical properties of silver and the morphology of the particles,[34] but these studies typically employ highly nonequilibrium shapes,[35−38] as the dramatic structural variations provide the greatest contrast in the optical response. Almost no attention has been given to the optical properties of silver NP at equilibrium, where the variations in shape are more subtle, partially because generating a set of equilibrium shapes as a function of size is (as stated above) extremely challenging. Even particles that appear to be spherical under low resolution have been shown to be faceted, and usually enclosed by combinations of low index surfaces,[39] the fraction of which may be sensitive to the surrounding conditions.[40,41] An alternative to the extensive experimental strategy needed to develop this understanding is to use theoretical and computational methods that can rapidly scan the configuration space with exquisite control.

In this paper we use a number of well-established modeling techniques to predict the equilibrium shape of silver NPs, as a function of diameter ($D$) and temperature ($T$), and generate a shape-dependent $\langle T, D\rangle$ phase diagram. For each the shapes occupying the $\langle T, D\rangle$ space we calculate the size-dependent optical properties and develop the first optical/structure property map of silver NPs at equilibrium. As we will show, spheres and faceted-spherical shapes such as the icosahedron, Marks decahedron, and truncated octahedron are the predominant morphologies in a Ag NPs colloidal sample at temperatures between 0 and 1200 K and with sizes smaller than 30 nm, producing solutions with a characteristic light yellow color. But, there is a small region at intermediate temperatures where the cube and tetrahedron are the stable geometries, and in consequence the sample shows a bright yellow hue and dark red color, respectively. These results indicate that over a broad range of $T$ and $D$ the surface observed color will be robust against structural variations and must be tuned by deliberately

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varying the concentration of the colloids or refractive index of the surrounding media.

## METHODOLOGY

In the present study the size- and temperature-dependent equilibrium shape of silver was modeled with a well-established thermodynamic model based on a geometric summation of the specific Gibbs free energy $G(T)$ of an entire NP[42] and includes the important phenomenological features such as the bulk, surfaces, and twins:

$$G(T) = \Delta G_f^\circ(T) + \frac{M}{\rho}\left(1 - \frac{2\sum_i f_i \sigma_i(T)}{B_0\langle R\rangle} + \frac{P_{ex}}{B_0}\right) \times [q\sum_i f_i\gamma_i(T) + n\sum_\theta (a_\theta \nu_\theta(T) + \sum_\phi l_{\theta\phi}\eta_\phi(T))] \quad (1)$$

where $M$ is the molar mass, $\rho$ is the mass density, $\gamma_i(T)$ is the temperature-dependent free energy of facet $i$, $\nu_\theta(T)$ is the energy of the twin plane of area $a$, in orientation $\theta$, and $\eta_\phi(T)$ is the re-entrant line tension where the twin plane intersects with the free surface in the orientation $\phi$, with length $l$. Dimensional consistency is preserved by introducing the number density of planar defects $n = n_t/V$ (where $n_t$ is the total number of defects and $V$ is the total volume). In this model $f_i$ is a weighting factor that defines the fractional surfaces areas, such that $\sum_i f_i = 1$. This model also accounts for the elastic effects of surface stress. The volume dilation induced by the isotropic surface stresses $\sigma_i$ and external pressure $P_{ex}$ is included using the Laplace–Young formalism using the bulk modulus $B_0$ and the average particle radius $\langle R\rangle$ calculated using a spherical approximation. In the following sections, atmospheric external pressure has been assumed ($P_{ex}$ = 101.33 kPa).

Briefly, it should ne noted that the full version of this model includes the contributions from edges and corners, as described in ref 42, but for such as large number of shapes the necessary parametrization can prove as computationally intensive as explicitly calculating the optimized geometry of complete, isolated nanostructures. Furthermore, the importance of edge and corner energies in the total free energy of the system is largely unknown. On the basis of the construction of the model, we can see that while $G_{surface} \propto 1/\langle R\rangle$, $G_{edge} \propto 1/\langle R\rangle^2$, and $G_{corner} \propto 1/\langle R\rangle^3$. This follows directly from the definition of the surface-to-volume, edge-to-volume, and corner-to-volume ratios ($q$, $p$, and $w$ in ref 42) and indicates that the contribution from the surfaces will dominate above a "critical size" that is to some degree independent of material. This issue was tested some years ago, and it was shown that total edge energies can be as much as 1–3 orders of magnitude smaller than the total energies for the adjacent surfaces under the critical diameter ($D_c$).[43] When considering nanostructures with a diameter less than $D_c$, it is still preferable to examine each morphology explicitly, by undertaking suitable calculations of isolated structures (using *ab initio* or tight-binding methods). The precise value of $D_c$ depends upon the material, but as a general rule of thumb $D_c \approx [(V\cdot 6 \times 10^4)/4\pi N]^{1/3}$, where $N$ is the number of atoms in the unit cell and $V$ is total volume of the unit cell. This consequently imposes a lower (size) limit for which accuracy may be assured and is why the truncated version of the model is not used below 4 nm and why the line tensions of the re-entrant edge *are* included for sizes <8 nm under well below the surface melting temperatures.

This version of the model requires the input of $B_0$, $\gamma_i(T)$, $\sigma_i(T)$, $\nu_\theta(T)$, and $\eta_\phi(T)$, which must be calculated explicitly for all facets $i$, $\theta$, and $\phi$ of interest, using (the same) appropriate computational method. In this case we have used density functional theory (DFT), within the generalized gradient approximation (GGA) the exchange-correlation functional Perdew–Burke–Ernzerhof (PBE),[44] with the projector augmented wave (PAW) potentials.[45,46] This has been implemented via the Vienna Ab initio Simulation Package (VASP)[47,48] which utilizes an iterative self-consistent scheme to solve the Kohn–Sham equations using an optimized charge-density mixing routine. Each model structure was fully relaxed, prior to the calculation of the total energies, using a 4 × 4 × 8 and 4 × 4 × 1 Monkhort–Pack $k$-point mesh, respectively. The electronic relaxation technique used herein is an efficient matrix-diagonalization routine based on a sequential band-by-band residual minimization method of single-electron energies,[49,50] with direct inversion in the iterative subspace, whereas the ionic relaxation involves minimization of the Hellmann–Feynman forces. The initial relaxations and the following static single point energy calculations were performed with a plane-wave basis up to 312 eV, and an energy convergence of $10^{-4}$ eV, including full spin polarization. This produces a bulk modulus of $B_0$ = 88.4 GPa (when $B_0' = 4$), computed with the Birch–Murnaghan equation of state.[52,53]

Since these DFT calculations have been performed at $T \approx 0$, a number of simple expressions have been used here to describe the temperature dependence. First, we have used a semi-empirical expression for determining of $\gamma_i(T)$ proposed by Guggenheim:[54]

$$\gamma_i(T) = \gamma_i(0)\left(1 - \frac{T}{T_c}\right)^a \quad (2)$$

where $a$ is an empirical parameter (known to be unity for metals[55]) and $T_c$ is the critical temperature at which the structure of the surface deteriorates or changes significantly from the structure in the bulk (is not longer fcc).[56] The value of $T_c$ is usually taken as the bulk surface melting temperature, but since this has been shown to be size dependent in metallic nanoparticles,[57] the expression of Qi and Wang[58] has also been employed:

$$T_c = T_m\left(1 - \frac{6r\Omega}{D}\right) = T_m(1 - qr) \quad (3)$$

where $T_m$ are the macroscopic surface melting temperatures, $D$ is the average diameter of the nanostructure, $r$ is the atomic radius of platinum, and $\Omega$ is a shape-dependent factor defined as the ratio of the surface area of the particle divided by the surface area of a sphere of equivalent volume.[58] Since $\sigma = \gamma + A\,\partial\gamma/\partial A$, the temperature dependence of $\sigma_i$ has been described in the same way. This procedure has been implemented with the surface melting temperatures from ref 59 and has been successfully used to produce phase diagrams for other metal NPs in the past.[60–62]

In general, the geometric parameters $\langle R\rangle$, $q$, $f_i$, $n$, $a_\theta$, and $l_{\theta,\phi}$ provide the size and shape dependence, and defining these parameters can be complicated, depending upon the shapes in question. At large sizes this model converges to the Wulff construction but has the advantage at small sizes of including additional influences that are important at the nanoscale.[63] More information on how to calculate $q$ and $f_i$ for fcc structures is provided in ref 64, and the 17 shapes investigated as part of

this study are shown in Figure 1. Many of these shapes are routinely observed in experiments, while others that are dissimilar to those observed are included for completeness and to perhaps elucidate why they are not routinely observed. It should be noted that although it is possible to apply these methods to anisotropic shapes, such as nanorods, this is considered to be beyond the scope of the present work.

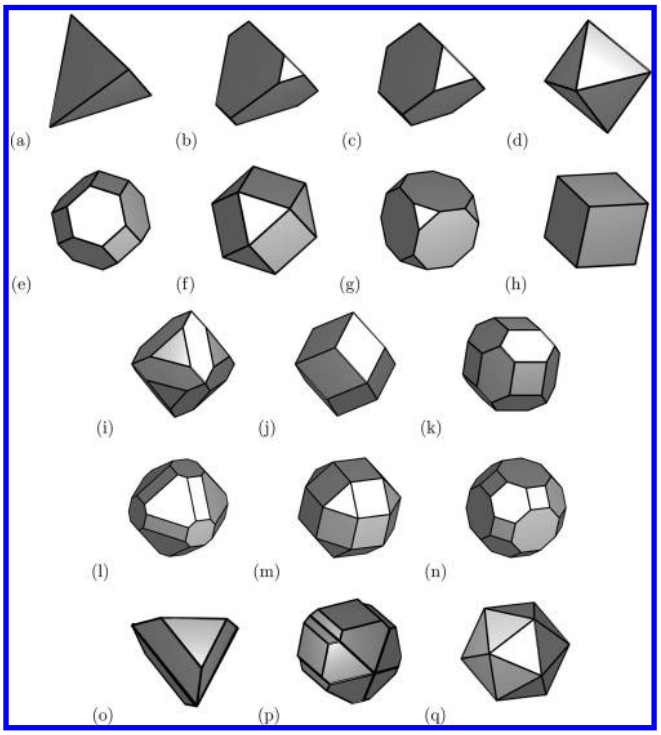


**Figure 1.** Silver nanoparticle morphologies: (a) tetrahedron, (b) minimally truncated tetrahedron, (c) truncated tetrahedron, (d) octahedron, (e) truncated octahedron, (f) cuboctahedron, (g) truncated cube, (h) cube, (i) truncated rhombic dodecahedron, (j) rhombic dodecahedron, (k) cubo-rhombic dodecahedron, (l) doubly truncated octahedron, (m) small rhombicuboctahedron, (n) great rhombicuboctahedron, (o) triangular prism, (p) Marks decahedron, and (q) regular icosahedron.

Once the size and shape of stable silver NPs are known, the solution of the Maxwell equations to the system of a single particle interacting with an electromagnetic (EM) field is possible. In consequence, the optical properties of a dilute system with a $c$ concentration of Ag NPs, all of them with the same size and shape, can be modeled. The temperature of the system is taken into account indirectly when considering the size of the NP and the corresponding stable shape at that temperature according to the $\langle T, D\rangle$ structure map. When the NP is a sphere the absorbed and scattered EM fields are calculated using the well-known solutions proposed by Mie,[27] but in the case of multifaceted shapes the discrete dipole approximation (DDA) is employed.[85] Once the extinction, absorption, and scattering efficiencies, $Q_{ext}$, $Q_{abs}$, and $Q_{sca}$, of a single NP are obtained, then the transmittance of the colloidal solution with a $c$ concentration is calculated using the Kubelka−Munk theory.[86] Later, as we explain below, the perceived color of the colloid by the human eye can be predicted. This methodology has been previously tested for isotropic and anisotropic Au NPs systems. The validity of the theory has been proved successfully when is compared to experimental results.[87,88]

The DDA is a method widely employed to study the optical properties of symmetric and asymmetric NPs. The fundamentals of the theory were introduced by Purcell and Pennypacker in ref 85. The main idea is to resemble the shape of the NP with an array of $N$ point dipoles. Given the position of each dipole and the dielectric function of the material at the wavelength of the incident field, then the problem is reduced to solve a set of $3N$ conjugated complex equations. We have solved numerically the system of equations using the DDSCAT CODE 7.3[93,94] which also calculates the optical efficiencies of the NP. Besides, the code allows to calculate the efficiencies considering size, shape, surrounding medium, and various orientations of the incident field. To guarantee the convergence of the involved calculations, we have considered a number of dipoles of the order of $10^5$; for example, a truncated octahedron was resembled considering $N$ = 162 521 and the cubo-rhombic dodecahedron with an $N$ = 136 003. Besides, we are considering that the NP is arbitrarily oriented in the colloidal solution, which means the optical response is the outcome of an average of orientations. In the case of morphologies with spherical symmetry, such as the icosahedron, truncated octahedron, and alike, due to its symmetry elements it was enough to consider only one orientation. Marks decahedron is a less symmetric morphology; it has a 5-fold symmetry axis. Then we considered three main orientations as in previous works.[82]

Regardless of the shape or symmetry, a key aspect to predict properly the optical properties of the NP is the dielectric function $\varepsilon$ used to describe the material. In general, the dielectric function of a metal depends on the wavelength of the incident field and is related to the electronic transitions of the free and bounded electrons named the intraband and interband transitions, respectively.[89] At the nanoscale these electronic transitions are size dependent. Whenever the radius of the NP is larger than 2 nm, it is possible to assume interband contributions independent of size.[90,91] On the other hand, intraband contributions are size dependent when the NP has a radius smaller than 50 nm; above this value size effects can be neglected, and therefore the bulk dielectric function can be used to estimate the optical properties.[92] Here, the predicted phase map of Ag NPs was calculated in the interval of diameters from 4 to 30 nm; therefore, we have considered size-independent interband contributions, and the intraband contributions have been modeled taking into account the damping of the free electrons due to the surface of the NP[95] as

$$\varepsilon(\omega, a) = \varepsilon_{bulk} + \frac{\omega_p^2}{\omega(\omega + i\gamma_{bulk})} - \frac{\omega_p^2}{\omega(\omega + i\gamma_{bulk} + iv_F/r)} \quad (4)$$

where $\omega$ is the frequency of the incident field, $\omega_p$ is the plasma frequency, $\gamma_{bulk}$ is the bulk damping term of the free electrons, $v_F$ is the Fermi velocity, and $r$ is the radius of the NP.

After the calculation of the optical efficiencies, $Q_{ext}$, $Q_{abs}$, $Q_{sca}$, the transmittance, $\mathcal{T}$, of a sample with a $c$ concentration of particles dispersed and randomly oriented in a medium with a refractive index, $m$, was calculated using the expression

$$\mathcal{T} = \frac{b}{a\sinh(bSL) + b\cosh(bSL)} \quad (5)$$

where $L$ is the thickness of the sample, $a = (S + K)/S$, and $b = (a^2 - 1)^{1/2}$ with $S$ and $K$ the Kubelka−Munk absorption and

scattering coefficients, respectively. Besides $S$ and $K$ are related to the optical efficiencies and NPs concentration as[86]

$$K = 2cQ_{abs}\pi r^2,\ S = 3Q_{sca}\pi r^2/4c \tag{6}$$

Finally, the color of a solution with a given concentration $c$ of Ag NPs, with an average radius $r$, and all of them with the same shape, which would be observed by the human eye, under transmittance was calculated choosing the $(L^*, a^*, b^*)$[86] color space and the power of a blackbody at $T$ = 6504 K (a daylight illuminant D65). Then, a color map of the phase diagram relating morphology and optical properties of the system at different temperatures is presented.

## ■ RESULTS AND DISCUSSION

**Phase Diagram.** Using the computational methods described above, the specific surface energies and isotropic surface stresses for the low index {111}, {110}, and {100} facets was calculated, using periodic slab structures with thickness of 26.2, 18.9, and 26.7 Å, respectively. This produced values of $\gamma_{(100)}$ = 768 mJ m$^{-2}$, $\gamma_{(110)}$ = 859 mJ m$^{-2}$, and $\gamma_{(111)}$ = 654 mJ m$^{-2}$ and $\sigma_{(100)}$ = −321 mJ m$^{-2}$, $\sigma_{(100)}$ = −315 mJ m$^{-2}$, and $\sigma_{(100)}$ = 477 mJ m$^{-2}$, respectively. The interfacial energy of $\nu_{\langle 111\rangle}$ = 219 mJ m$^{-2}$ was determined using 1 × 1 periodic cell of 56 Å thickness perpendicular to the twin plane, and the line tensions of $\eta_{\langle 100\rangle} = -1.035 \times 10^{-7}$ mJ m$^{-1}$ and $\eta_{\langle 111\rangle} = -4.255 \times 10^{-7}$ mJ m$^{-1}$ were fitted to cluster calculations (using the same computational settings as above), as described in ref 42. The fact that the line tensions are exothermic is important in the stabilization of twinning at small sizes, as we previously shown for gold.[42]

Using these input parameters, and the theoretical methods described and referenced above, the stability of each of the silver nanoparticle morphologies was predicted. By comparing these predictions in the ranges 4−30 nm and 0−1500 K, the size- and temperature-dependent phase diagram of silver nanoparticles was constructed, mapping the lowest energy morphology in the ⟨$T$, $D$⟩ manifold, as shown in Figure 2.

We can see from Figure 2 that at low temperatures we have as size-dependent energetic ordering from icosahedra (with a high fraction of planar defects), to a Marks decahedron (with a moderate fraction of planar defects), to pristine truncated octahedra at large sizes, which are consistent with the bulk Wulff construction. This trend is consistent with all previous theoretical results: that Ag small icosahedra, intermediate-size decahedra, and large truncated octahedra clusters are expected at low temperature.[65,66] These results are also in exceptionally good agreement with the previously reported phase diagram for silver clusters at room temperature,[67] even though it was based on empirical Sutton−Chen[68] potentials rather than DFT.

This is also consistent with experimental observations of Ag nanoparticles <5.5 nm grown at low temperature with inert gas aggregation.[69] Although larger icosahedra are often observed, in regions where fcc structures are expected, it was established many years ago that these unstable (or metastable) structures were due to kinetic factors and that the equilibrium shape could be recovered by raising the temperature.[5,60,80] The same interplay between thermodynamic and kinetic influences has been observed, and modeled, in gold.[51,71−78]

In some ways this result is similar to the phase diagram for gold,[60] as in both cases the exothermic re-entrant line tensions compensate for the endothermic contribution to the total free energy from the twin boundary plane and certain configurations

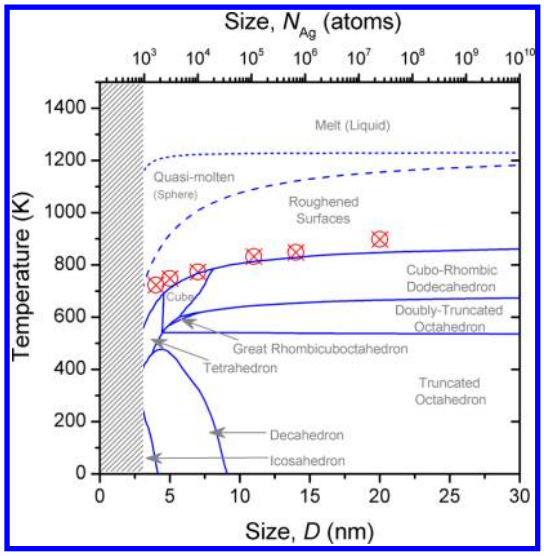


**Figure 2.** Size- and temperature-dependent phase diagram of silver nanoparticles, mapping the lowest energy morphology in the ⟨$T$, $D$⟩ manifold. The ⊗ symbols indicate the experimental observations of the melting temperature by Asoro et al., where surface effects were expected to dominate,[81] and the shaded area is beneath the range of applicability of the model.

stabilize the defect. This is not the case in Pd[62] or Pt,[61] where the re-entrant line tensions are endothermic, and so twinning is far less stable.

Above room temperature, however, we start to see the emergence of other morphologies that have not been included in previous studies. At small sizes, just below the surface roughening temperatures, we see a competition between the cube and tetrahedron. Both shapes have a high surface-to-volume ratio, and as the temperature dependence significantly lowers the surface energy (with respect to the bulk) just before roughening, shapes that maximize the surface-to-volume ratio become most favorable. Below 5 nm this size dependence dominates, but as as the size increases, the cube gains in relative stability due to the lower surface melting temperature.[59] At large sizes, as we converge to the bulk Wulff shape, at the given temperature, and favorability of high surface-to-volume ratios rapidly decays. In this region we see the emergence of the {110} facets in addition to the {100} facets—again due to the substantial lowering of the surface energy by the lower surface melting temperatures. This commences at ∼540 K with the doubly truncated octahedron, which has previously been observed in supported Ag nanoparticles, synthesized by burning magnesium ribbons wrapped in Ag wires at temperatures sufficiently high to equilibrate the shape.[79] A higher proportion of {110} facets has been achieved experimentally via sublimation-induced equilibration of silver cubes at around ∼1000−1100 K, which is also in excellent agreement.

As we raise the temperature further, we see the onset of surface roughening, as the first of the (size-dependent) surface melting temperatures is reached, and then a quasi-molten (amorphous) structure before the entire particle melts (liquid). Figure 2 also contains the experimental observations of the melting temperature by Asoro et al., where surface effects were expected to dominate.[81]

While this high temperature behavior is expected, it is interesting to see when that at a number of morphologies compared herein do not participate in the phase diagram, at any

point in ⟨$T$, $D$⟩ space. This including a number of morphologies enclosed by the a considerable fraction of {111} surface, known to be most stable at low temperature, such as the cuboctahedron, the truncated rhombic dodecahedron, the small rhombicuboctahedron, the different truncated tetrahedra, and the triangular prism. In particular, the triangular prism and similar shapes are often observed in experiments,[37] and so based on these findings, it is reasonable to suggest this shape could be due to kinetic influences. If we once again look to gold for insights, we can see that rapid coarsening in specific crystallography directions can drive structures out of equilibrium, as can the combination of thermodynamic and kinetic influences acting together.[51] Alternatively, it could be that, since experimental samples are never truly monodispersed, there is simply a statistically significant chance that some non-equilibrium shapes will be exist, regardless of the care taken during synthesis.

**Optical Spectrum.** According to the phase map is possible to govern the morphology of the NP by following one of two options: (a) fixing the temperature and varying the size or (b) fixing the size and changing the temperature. Any of these two options would be the right one depending on the ambient conditions and possible applications. Foremost, we have focus on option (a) because several methods allow to grow silver NPs controlling its size by changing the amount of silver salts needed for the chemical reactions. In this sense and with illustrative purposes we present in Figure 3a the optical

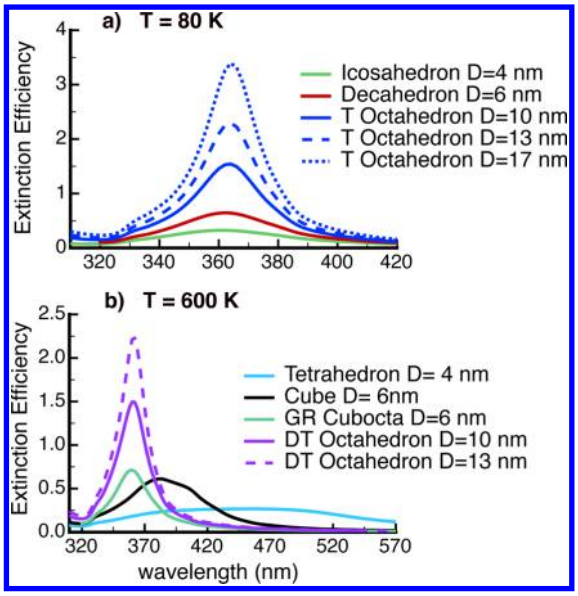


**Figure 3.** Optical response of Ag NPs with different sizes and morphologies at a temperature (a) $T$ = 80 K and (b) $T$ = 600 K.

response of silver NPs at a temperate $T$ = 80 K and with different sizes, therefore with different morphologies, according to Figure 2. At this particular value of temperature an icosahedral structure is possible whenever the size of the NP is smaller than 4 nm, the decahedral morphology is stable in the interval from 4 to 9 nm, and above this value a truncated octahedra is stable. These morphologies have a spherical-like symmetry, and we observe that all the spectra are wide and have a dominant peak at around 360 nm. At this interval of diameters, between 4 and 17 nm, the size does not alter considerably the position of the resonance because radiation damping is negligible, as is observed from the spectra of the truncated octahedra of 10, 13, and 17 nm. The differences in intensity are a consequence of the proportionality of the optical efficiency to the volume of the NP. Besides, size effects also have an influence over the intensity because surface damping becomes more relevant as the size of the NP diminishes. Otherwise, the full width at half-maximum (FWHM) can be used to define the width of the spectra, and we found that these values are 56, 44, 30, 26, and 24 nm for $D$ = 4, 6, 10, 13, and 17 nm, respectively. Then, the trend is that the FWHM decreases as the size increases, mainly due to surface damping. Because all these morphologies have a spherical symmetry, the width of its spectrum is due to the presence of more than one SPRs with similar intensities and with resonance frequencies close to each other. The position and number of SPRs of each morphology can be estimated considering the nondissipation limit, i.e., not taking into account dissipation effects due to size effects and damping terms ($\gamma_{bulk} \rightarrow \infty$) of free electrons.[82]

On the other hand, at temperatures above room temperature the optical properties are strongly determined by the morphology and the size, as is shown in Figure 3b. Here we show the efficiencies of silver NPs at 600 K with diameter from 4 to 13 nm. At this temperature the stable morphologies are the tetrahedron (4 facets), cube (6 facets), great rhombi (GR)-cuboctahedron (26 facets), and double truncated (DT) octahedron (26 facets). The last two shapes have a spherical-like symmetry, and their spectra show a main SPR at 360 nm. Therefore, the shape line and SPR of their spectra can be understood keeping in mind the same arguments that explain Figure 3a. On the basis of the structural map (Figure 2), we notice the GR-cuboctahedron is stable in a very tiny interval of sizes and temperatures, and then it would be difficult to find this shape in experimental samples. The cube and the tetrahedron are less symmetric and are characterized by acute vertices which produce wide spectra covering from the blue to the red, indicating the presence of several surface plasmons of high multipolar order.[83,84] We observe that, at least for this temperature value, as the size of the NP increases, the number of facets increases and the shape tends to a sphere. Therefore, the transition from a small tetrahedral NP to a medium size DT octahedron is represented by a wide spectrum shifting to the blue and giving place to an intense and slight peak due to the degeneration of the resonances. We also need to remark that experimental groups have synthesized silver cubes with side length larger than 90 nm,[96] but this requires deliberate intervention by the experimentalist and enlists kinetic driving forces. The size/temperature/morphology relationship is more complex when we depart from equilibrium, and so the trend observed in the structural map should not be extrapolated to diameters larger than 30 nm unless the synthesis conditions are known to be thermodynamically controlled.

In Figure 4, we illustrate how an increase of the ambient temperature where the NP is at thermal equilibrium promotes changes in its morphology and therefore in its optical efficiency. As the temperature increases from 0 to 1200 K, before melting, the Ag NP goes through several morphology transitions, but the interval of temperature of each transition depends on the size of the NP. To expose this, we have chosen three representative cases: a NP with a diameter $D$ = 4 nm, another with $D$ = 6 nm, and one last with $D$ = 17 nm, represented in parts a, b, and c of Figure 4, respectively. A general behavior of NPs with a diameter smaller than 8 nm indicates that at low temperatures silver atoms prefer an array forming faceted-spherical shapes, such as the icosahedron with 20 facets or the

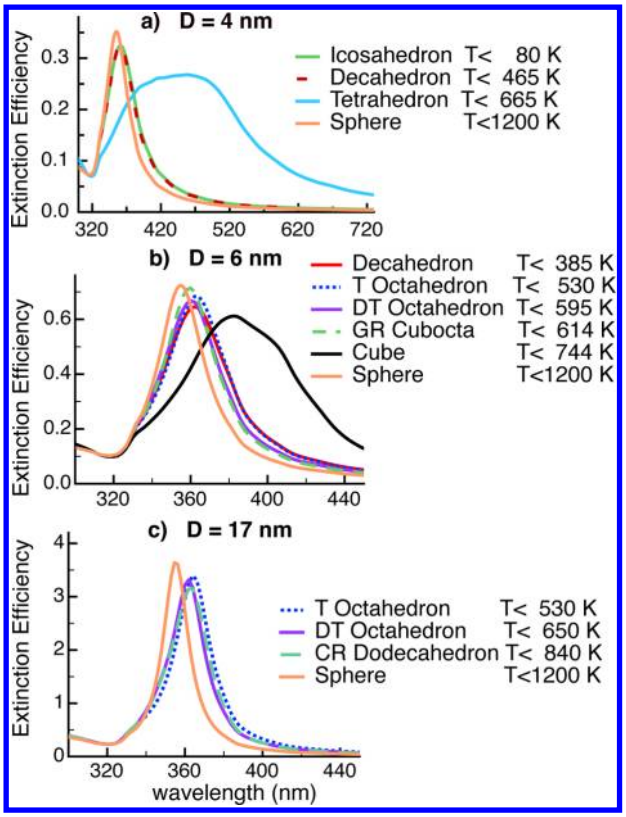


**Figure 4.** Variations in the optical response of Ag NPs of diameter (a) 4, (b) 6, and (c) 17 nm as the ambient temperature increases.

DT octahedron with 26 facets. Before becoming a sphere, which occurs at high temperatures, the preferred shape is still faceted but defined by acute vertices, either a cube or a tetrahedron. Then, although at low and high temperatures the spectrum is narrow and located in the violet-blue region, at intermediate temperatures the spectrum is wide covering from the blue to the red region of the visible spectrum.

The shape transitions of NPs with a diameter between 10 and 30 nm involves only faceted-spherical shapes that includes the truncated (T) octahedron, DT octahedron, and cubo-rhombic (CR) dodecahedron. Figure 4c shows the variations of the optical response according to the shape transitions that take place in a Ag NP with a diameter of 17 nm as it is heated up from 0 to 1200 K. Because of the spherical-like symmetry of the respective shapes, the spectra are narrow and located in the violet-blue region.

**Optical Structure/Property Map.** Once the transmittance of a colloid sample is calculated using the optical efficiencies of Ag NPs with a *c* concentration, then the color of the sample appreciated by the human eye can be estimated following the methodology explained above in the Methodology section. Figure 5 shows the color map of a colloid of Ag NPs at different sizes and temperatures. We have assumed the same volume concentration for all the samples, which means the number of NPs in a sample with diameter 2 nm is larger than in the sample with NPs with diameter 20 nm, but the amount of silver is the same. As a reference, we indicate that the sample with NPs with a diameter of 2 nm has a concentration of $8 \times 10^{13}$ particles/mL. For the whole range of sizes and temperatures of the calculated color map we appreciate the same color, light yellow, except in the region where the cube and tetrahedron dominate where a bright yellow hue and a dark red color are

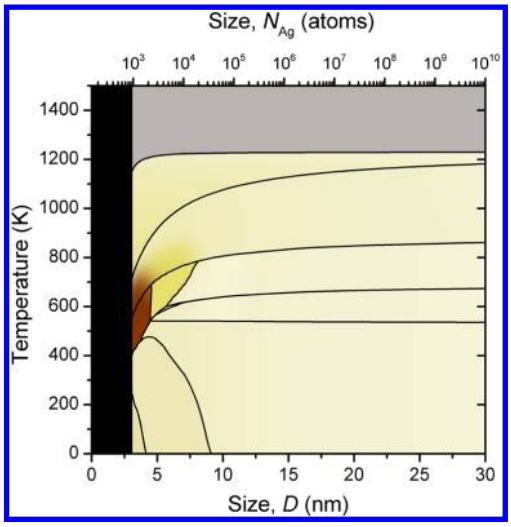


**Figure 5.** Structure/optical map of a colloidal sample of Ag NPs. The color has been calculated considering the same volume concentration for all the samples.

observed. This is not unexpected because of the similarities between the spectra of the faceted-spherical shapes and the differences with the spectra of the cube and tetrahedron.

These results may suggest that opportunities of tuning the optical spectrum of Ag NPs are limited, but that is not the case. It is possible to produce nonequilibrium shapes that will have a different characteristic spectrum, though they will be unstable with respect to shape changes (to one represented on the phase diagram) given a suitable perturbation. In addition to this, the size and shape of individual NPs are not the only experimental parameters that may be tuned. Chemical synthesis methods allow to grow Ag NPs within solutions of water, toluene, or alcohol and with various particle concentrations, which produce a variety of hues perceived by the naked eye. As we can see from Figure 6, by increasing the concentration of the colloids, a

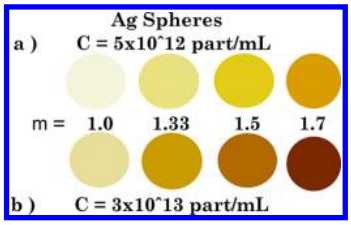


**Figure 6.** Colloid solution of spherical Ag NPs with a diameter 10 nm at two different concentrations and within surrounding medium with refractive index $m$ = 1 (vacuum), 1.33 (water), 1.5 (toluene), and 1.7 (sapphire).

darker color can be obtained. The hue can also be tuned by varying the refractive index of the surrounding solution. These are far more convenient methods for tuning the surface plasmons, as they do not require complicated synthesis procedures and can be altered (postsynthesis) as often as required.

## ■ CONCLUSIONS

Therefore, although considerable effort and investment are focused on delivering silver nanoparticles with precise morphologies, we find that the most stable (equilibrium shapes) have a remarkably similar optical emission. Using a

combination of computational and theoretical approaches, we have constructed a size- and shape-dependent phase diagram of nanoscale silver and find that faceted near-spherical shapes and spheres are the predominant shapes in a Ag NPs colloidal sample at low and high temperatures. By then projecting the optical emission spectrum to produce a detailed structure/property map, we show that a light yellow color may be expected when samples are at thermal equilibrium, with the exception of cubic shapes such as cubes or tetrahedrons which present brighter yellow hues and dark red colors at small sizes and intermediate temperatures. These results indicate that over a broad range of $T$ and $D$ the surface observed color will be robust against structural variations and must therefore be tuned by deliberately varying the concentration of the colloids or refractive index of the surrounding media. An advantage of this (lately) insensitivity of the optical emission with the size and shape of particles at equilibrium is that once the low-energy structure has been achieved, color variations will become a sensitive indicator of aggregation (which changes the local concentration) or changes in the refractive index of the local environment, or possible oxidation of the silver NP.

The production of the shape-dependent phase diagram of silver nanoparticle also has implications for other areas of research, beyond the optical properties.[97−100] Recent studies have demonstrated that there is a close relation between the shape of silver nanoparticles and their antibacterial and antiviral properties,[101] particularly where the fraction of {111} facets varied.[102,103] In the future it will be possible to use the phase diagram presented here to generate structure/property maps for applications such as these, once all the details of the size and shape dependence have been confirmed.

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### Notes

The authors declare no competing financial interest.

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